Name → Eesh Saxena

GITHUB → github/eeshsaxena/cs8081-cloud-mini-project.

What I learnt →

How sequence-level knowledge works - training a small student (pythia-70m) on a teacher's (phi-3.5) generated text, not its weights; logit-level was not possible as Ollama returns only text.

I also learnt object-storage (seaweed FS/S3), running kubernetes with k3d (probes, resources, limits, HPA, rolling updates), stateless NACL rule ordering, and CI with a quality gate.

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Challenges and Solutions →

→ Teacher added prose after its JSON (17% valid) → used Ollama format: "json". 100% valid.

→ Student scored 0% → a trailing space broke tokenization; ending it as "JSON". fixed pt (0 → 99%)

→ My laptop could not run Docker (no WSL2 / virtualization) → ran the k3d cluster on GitHub Actions

→ kept the image ≤ 2GB with CPU-only Pytorch, weights as Release assets, not in git.

~~Findings~~ →

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Findings →

Student full frame 43% vs teachears 48%, 99% valid JSON - a 70M model underfits. Cluster: image 1.74 GB, cold start 8s, p95 200ms at 2.41 rps, load balanced 51/49, SLO 99.1%, ~0.91 INR per 1000 requests, NACL 6/6.

Improvements →

Requests batching, more epochs/a larger students, a shared work queues, a multi-node cluster with a PodDisruptionBudget, and per-service scoped S3 keys.

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